A sharp objective bound and a residual counterexample for Douglas–Rachford splitting

Abstract

For smooth-first unrelaxed Douglas–Rachford splitting, we prove the exact terminal objective bound $\|w^1 - w^\star\|^2/(4\gamma N)$ whenever the convex smooth term has Lipschitz gradient constant L and $0 < \gamma L \le 1$. The second term may be any proper closed convex function with a compatible primal–dual solution. This establishes the unrelaxed small-step part of a recent composite-objective conjecture. The proof expands an explicit weighted interpolation certificate into nonnegative squares for every horizon. A classical proximal-point example attains the constant. For a separate operator-residual conjecture in the same source, we give a two-dimensional rational cocoercive counterexample at horizon fourteen and a family with squared residual asymptotic to $1/(5eN)$ at its chosen horizon. The latter examples use the established averaged-rotation mechanism and have no convex-gradient representation.

Keywords: Douglas–Rachford splitting; exact objective bound; convex interpolation; cocoercive operator

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1 The objective bound and its scope

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex with L-Lipschitz gradient, where $L > 0$, and let $g : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be proper, closed and convex. Write $F = f + g$. Assume that

$$p^\star = \nabla f(x^\star), \qquad -p^\star \in \partial g(x^\star), \qquad w^\star = x^\star + \gamma p^\star. \tag{1}$$

The smooth-first unrelaxed Douglas–Rachford iteration is

$$\begin{aligned} x^k &= \operatorname{prox}_{\gamma f}(w^k), \\ y^k &= \operatorname{prox}_{\gamma g}(2x^k - w^k), \\ w^{k+1} &= w^k + y^k - x^k, \qquad k = 1, 2, \dots. \end{aligned} \tag{2}$$

The initial state w^1 is arbitrary. The reported point y^N is obtained after $N - 1$ state updates and the Nth pair of proximal evaluations.

Theorem 1. *If $0 < \gamma L \le 1$, then for every integer $N \ge 1$ the actual iterates (2) satisfy*

$$F(y^N) - F(x^\star) \le \frac{\|w^1 - w^\star\|^2}{4\gamma N}. \tag{3}$$

For every admissible γ, L, N and every positive initial radius, this constant is attained by finite-valued convex functions.

Conjecture 2 in the version dated 13 September 2025 of Abbaszadehpeivasti–Zamani [1] proposes, for the relaxed update $w^{k+1} = w^k + \lambda(y^k - x^k)$,

$$F(y^N) - F(x^\star) \le \frac{\|w^1 - w^\star\|^2}{4\gamma((N-1)\lambda + 1)}$$

throughout $0 < \gamma L < 2\sqrt{2} - 1$ and $0 < \lambda < (1 + \sqrt{5})/2$. Theorem 1 proves the part with $\lambda = 1$ and $\gamma L \le 1$. The larger stepsize and other relaxation ranges remain outside the result. The cited v2 has numerator 1; its v1 printed numerator L. The source's Section 5 literally declares both functions finite-valued. Our proof also permits extended-valued g, for which every second shadow still belongs to $\operatorname{dom} g$.

The nearest general terminal-shadow theorem of Davis–Yin [2, Theorem 3.2] gives

$$F(y^N) - F(x^\star) \le \frac{\|\operatorname{prox}_{\gamma f}(w^1) - x^\star\|^2}{2\gamma N}, \qquad \gamma L < \kappa \simeq 1.24698,$$

where κ is the positive root of $\kappa^3 + \kappa^2 - 2\kappa - 1 = 0$. Nonexpansiveness of the first prox bounds its numerator by $\|w^1 - w^\star\|^2$. Thus (3) improves the uniform coefficient in this common state-radius normalization; the two original numerators can differ on a particular instance. The exact proximal-point objective coefficient and matching norm-function examples are already known; see Wang et al. [3, Theorem 3.1 and Lemma 2.3]. The contribution here is the upper bound for arbitrary coupled smooth f and convex g. Existing convex-envelope transfers assume a quadratic smooth term [4]. The exact ADMM dual-gap theorem [5, Theorem 3] concerns a curvature-bearing first primal block and a different dual shadow. A consistent initialization can control its product-space radius by a DR-state radius, but it does not supply the required curvature, order and terminal-shadow alignment.

The other assertion considered below is an operator-residual conjecture from [1]. It has different hypotheses and a different observable. Sections 5–6 treat it separately, using classical rotation examples with explicit attribution.

2 A final interpolation certificate

Translate $x^\star$ to zero and define

$$H(u) = \gamma\big(f(x^\star + u) - f(x^\star) - \langle p^\star, u\rangle\big),$$
$$K(u) = \gamma\big(g(x^\star + u) - g(x^\star) + \langle p^\star, u\rangle\big).$$

Then $H(0) = K(0) = 0$, $\nabla H(0) = 0$, $0 \in \partial K(0)$, and H has gradient Lipschitz constant at most one. Put $W_i = w^i - w^\star$, $X_i = x^i - x^\star$, $Y_i = y^i - x^\star$ and

$$P_i = \nabla H(X_i), \quad Q_i = X_i - P_i - Y_i \in \partial K(Y_i), \quad D_i = P_i + Q_i, \quad C = \nabla H(Y_N).$$

The actual recurrence becomes

$$X_i = W_i - P_i, \qquad Y_i = W_i - D_i - P_i, \qquad W_{i+1} = W_i - D_i. \tag{4}$$

All sampled function values are finite. Proximal well-posedness follows from proper closed convexity.

For sampled smooth points define

$$I_H(U,V) = H(U) - H(V) - \langle \nabla H(V), U - V\rangle - \tfrac{1}{2}\|\nabla H(U) - \nabla H(V)\|^2 \geq 0.$$

For sampled nonsmooth points, using their actual subgradients, define

$$I_K(U,V) = K(U) - K(V) - \langle Q_V, U - V\rangle \geq 0, \qquad Q_0 = 0.$$

The first inequality is the standard smooth convex interpolation inequality; it also follows by applying unit-step descent to $H - \langle \nabla H(V), \cdot\rangle$, which is minimized at V. Its coefficient remains valid when the actual smoothness is smaller than one.

Fix $N \geq 1$ and set

$$\begin{aligned}
\sigma_i &= \frac{i}{2N-i} \quad (0 \leq i \leq N), \qquad a_i = \sigma_i - \sigma_{i-1} = \frac{2N}{(2N-i)(2N-i+1)},\\
E_i &= \frac{2i}{2N-i+1}, \qquad F_i = \frac{i}{2N-i} \quad (1 \leq i < N),\\
A_i &= \frac{2[2N^2 - (N+1)i]}{(2N-i)(2N-i+1)(2N-i+2)} \quad (1 \leq i < N),\\
A_N &= \frac{2(2N+1)}{(N+1)(N+2)}, \qquad \theta = \frac{1}{N+1}, \qquad \rho = \frac{N}{N+1}.
\end{aligned} \tag{5}$$

Let $E_0 = F_0 = 0$. Every coefficient in (6) is positive on its nonempty index range. In particular $2N^2 - (N+1)i \geq N^2 + 1$ for $i < N$.

Define the weighted sum

$$\begin{aligned}
\mathcal{I}_N = &\sum_{i=1}^{N} A_i I_H(0, X_i) + \theta I_H(X_N, 0) + \theta I_H(0, Y_N) + \rho I_H(X_N, Y_N)\\
&+ \sum_{i=1}^{N-1} \big(E_i I_H(X_i, X_{i+1}) + F_i I_H(X_{i+1}, X_i)\big)\\
&+ \sum_{i=1}^{N} a_i I_K(0, Y_i) + \sum_{i=1}^{N-1} \sigma_i I_K(Y_i, Y_{i+1}).
\end{aligned} \tag{6}$$

Empty sums vanish. The final quadratic certificate is

$$\mathcal{R}_N = \frac{\|W_1\|^2}{4N} - H(Y_N) - K(Y_N) - \mathcal{I}_N. \tag{7}$$

Lemma 1. *Substituting* (4) *cancels every function value in* (7)*. The resulting quadratic form is nonnegative for all freely chosen vectors* $(W_1, D_1, \ldots, D_N, P_1, \ldots, P_N, C)$ *and every* $N \geq 1$*.*

The cancellation follows from

$$A_i = (E_i - F_i) - (E_{i-1} - F_{i-1})\ (i < N), \quad A_N = 1 - E_{N-1} + F_{N-1}, \quad \theta + \rho = 1,$$

and $a_i = \sigma_i - \sigma_{i-1}$. The nonnegativity is proved in the next section. Since $\mathcal{I}_N \geq 0$ on actual iterates, the lemma gives the normalized upper bound. Dividing by γ proves (3).

3 An all-horizon sum of squares

We prove Lemma 1. Quadratic matrix notation below is interpreted through inner products, so the scalar identities apply in every finite dimension. Put $z = (P_1, \ldots, P_N, C)$. Define a symmetric tridiagonal matrix B by

$$\begin{aligned} B_{ii} &= 2A_i + 2E_{i-1} + \sigma_i && (i < N), \\ B_{NN} &= 1 + 2\sigma_{N-1}, \qquad B_{N+1,N+1} = \tfrac{1}{2}, \\ B_{i,i+1} &= -E_i - \tfrac{1}{2}\sigma_i && (i < N), \\ B_{N,N+1} &= -\frac{N-1}{2(N+1)}. \end{aligned} \tag{8}$$

Unlisted entries are zero. Direct expansion of (6) gives

$$\begin{aligned} \mathcal{R}_N = \mathcal{R}_0 &+ z^T B z + \sum_{i=1}^{N} (a_i - A_i)\langle P_i, W_i\rangle \\ &+ \sum_{i<N} \langle E_i P_{i+1} - \sigma_i P_i, D_i\rangle - \sum_{i=2}^{N} \sigma_{i-1}\langle P_{i-1}, D_i\rangle + \langle C, D_N - \theta W_N\rangle, \qquad (9) \\ \mathcal{R}_0 &= \frac{\|W_1\|^2}{4N} - \sum_{i=1}^{N} a_i \langle D_i, W_i\rangle + \sum_{i=1}^{N} \sigma_i \|D_i\|^2. \end{aligned}$$

For example, the adjacent differences used in this expansion are

$$X_i - X_{i+1} = D_i - P_i + P_{i+1}, \qquad Y_i - Y_{i+1} = D_{i+1} - P_i + P_{i+1}.$$

The forward and reverse smooth slacks contribute

$$\langle E_i P_{i+1} - F_i P_i, X_i - X_{i+1}\rangle + \tfrac{1}{2}(E_i + F_i)\|P_i - P_{i+1}\|^2$$

to the remainder. The nonsmooth terms contribute

$$-a_i\langle D_i - P_i, W_i - D_i - P_i\rangle + \sigma_i\langle D_{i+1} - P_{i+1}, D_{i+1} - P_i + P_{i+1}\rangle,$$

with the second term only for $i < N$. These identities, together with the terminal terms, verify (9).

3.1 Completing the state squares

For $1 \le i \le N+1$ let $m_i = 2N - i + 1$, and for $1 \le i \le N$ put

$$\kappa_i = N(2i-1) - i(i-1) = N + (i-1)(2N-i) > 0, \qquad r_i = \frac{W_i}{m_i} - \frac{W_{i+1}}{m_{i+1}}. \tag{10}$$

Then

$$\mathcal{R}_0 = \sum_{i=1}^{N} \kappa_i \|r_i\|^2. \tag{11}$$

Expansion of $\mathcal{R}_0$ in the state coordinates gives coefficient $1/(4N)$ for $\|W_1\|^2$, coefficient $2\sigma_{i-1}$ for $\|W_i\|^2$ when $2 \le i \le N$, and coefficient 1 for $\|W_{N+1}\|^2$. The coefficient of $\langle W_i, W_{i+1}\rangle$ is $-\sigma_i - \sigma_{i-1}$. These agree with (11) because

$$\kappa_1 = N, \quad \kappa_N = N^2, \quad \kappa_{i-1} + \kappa_i = 2(i-1)m_i, \quad \frac{\kappa_i}{m_i m_{i+1}} = \frac{\sigma_i + \sigma_{i-1}}{2}.$$

Set $P_0 = 0$, $\beta_0 = 0$, and

$$\beta_i = \frac{i(2N-i-1)}{2N-i} \quad (1 \le i < N).$$

Define the vector linear forms

$$T_i = 2iP_{i+1} - \beta_i P_i - \beta_{i-1} P_{i-1} \quad (i < N), \qquad T_N = NC - \beta_{N-1} P_{N-1}. \tag{12}$$

The cross terms in (9) equal $\sum_i \langle T_i, r_i\rangle$. This follows by collecting each P_i and C coefficient and using

$$(a_i - A_i)m_i = 2\sigma_i - E_{i-1} \ (i < N), \qquad (a_N - A_N)m_N = -E_{N-1},$$

$$m_{i-1}E_{i-1} = 2(i-1), \qquad \beta_i = m_{i+2}\sigma_i, \qquad D_N - \theta W_N = N r_N.$$

Thus, if t_i is the coefficient row of T_i in z,

$$\mathcal{R}_N = \sum_{i=1}^{N} \kappa_i \left\| r_i + \frac{T_i}{2\kappa_i} \right\|^2 + z^T G z, \qquad G = B - \sum_{i=1}^{N} \frac{t_i^T t_i}{4\kappa_i}. \tag{13}$$

It remains to prove $G \succeq 0$.

3.2 The remaining band matrix

First suppose $N \ge 4$. The matrix G is pentadiagonal. Put

$$v_i = \frac{(i+1)\beta_i}{2\kappa_{i+1}} \quad (1 \le i \le N-2), \qquad v_{N-1} = \frac{N-1}{4(N+1)}, \qquad v_0 = v_N = 0. \tag{14}$$

The second off-diagonal satisfies $G_{i,i+2} = v_i$. Define

$$e_i = -G_{i,i+1} - 2(v_{i-1} + v_i) \quad (1 \le i \le N), \qquad \eta_i = \sum_{j=1}^{N+1} G_{ij}.$$

There is the exact identity

$$\begin{aligned} z^T G z = & \sum_{i=1}^{N-1} v_i \|z_i - 2z_{i+1} + z_{i+2}\|^2 \\ & + \sum_{i=1}^{N} e_i \|z_i - z_{i+1}\|^2 + \sum_{i=1}^{N+1} \eta_i \|z_i\|^2. \end{aligned} \tag{15}$$

Here $z_i = P_i$ for $i \le N$ and $z_{N+1} = C$. The second off-diagonals, first off-diagonals and row sums agree on both sides; these determine a symmetric pentadiagonal matrix. The weights v_i are positive for $1 \le i < N$.

For $1 \le i \le N-2$, put $t = 2N - i$. Simplifying (13) gives

$$e_i = \frac{i\,P(i,t)}{4t\kappa_i\kappa_{i+1}},$$

where

$$2P(i,t) = (2i^2 - i + 2)t^2 + (3i^2 + 3i - 2)(t-1) + (2i^2 + 5i - 2) > 0.$$

Each coefficient is positive for $i \ge 1$, and $t \ge 1$. The terminal weights are

$$e_{N-1} = \frac{(N-1)^2(N+2)}{2(N+1)(N^2-2)} > 0, \qquad e_N = 0. \tag{16}$$

The boundary row sums are

$$\begin{aligned} \eta_1 &= \frac{(N-1)(7N-4)}{2N(2N-1)(3N-2)}, \\ \eta_{N-1} &= \frac{N(3N^5 - 4N^4 - 27N^3 + 18N^2 + 30N - 28)}{4(N+1)(N+2)(N+3)(N^2-6)(N^2-2)}, \\ \eta_N &= \frac{5N^2 - N - 10}{2(N+2)(N^2-2)}, \qquad \eta_{N+1} = \frac{1}{2(N+1)}. \end{aligned} \tag{17}$$

They are positive for $N \ge 4$. For the only non-immediate numerator, set $q = N - 4 \ge 0$; its expansion is

$$3q^6 + 68q^5 + 613q^4 + 2786q^3 + 6614q^2 + 7476q + 2800 > 0.$$

For $2 \le i \le N-2$, the interior row sum is

$$\eta_i = \frac{Q(i-2, N-i-2)}{2t(t+1)(t+2)\kappa_{i-1}\kappa_i\kappa_{i+1}} > 0, \qquad t = 2N - i, \tag{18}$$

where the explicit polynomial Q in Appendix A has positive coefficients and a positive constant term. All denominator factors are positive. The appendix also gives a local rational identity deriving these row sums from (13). This proves $G \succeq 0$ for every $N \ge 4$.

3.3 The three small horizons

At $N = 1$, direct expansion gives

$$\mathcal{R}_1 = \|W_1/2 - D_1 - C/2\|^2 + \|P_1\|^2 + \|C\|^2/4 \geq 0.$$

At $N = 2$, the matrix in (13) satisfies

$$\begin{aligned} z^T G z = {} & \tfrac{1}{12}\|P_1 - 2P_2 + C\|^2 + \tfrac{1}{3}\|P_1 - P_2\|^2 \\ & + \tfrac{1}{6}\|P_1\|^2 + \tfrac{1}{2}\|P_2\|^2 + \tfrac{1}{6}\|C\|^2. \end{aligned}$$

At $N = 3$, it satisfies

$$\begin{aligned} z^T G z = {} & \tfrac{4}{35}\|P_1 - 2P_2 + P_3\|^2 + \tfrac{1}{8}\|P_2 - 2P_3 + C\|^2 \\ & + \tfrac{29}{105}\|P_1 - \tfrac{12}{29}P_2\|^2 + \tfrac{9607}{24360}\|P_2 - \tfrac{8700}{9607}P_3\|^2 \\ & + \tfrac{47157}{96070}\|P_3\|^2 + \tfrac{1}{8}\|C\|^2. \end{aligned} \tag{19}$$

These are exact square identities for (13). The separate treatment of $N = 3$ matters: its unmodified row sum $\eta_2 = -5/168$ is negative, so the argument for $N \geq 4$ does not apply to it. Equations (13)–(19) prove Lemma 1 in all horizons.

4 Attainment of the objective bound

Fix $R > 0$ and take

$$f = 0, \qquad g(u) = \frac{R|u|}{2\gamma N}, \qquad w^1 = R, \qquad w^\star = x^\star = 0.$$

This is the standard norm-function lower example for proximal point [3]. Let $h = R/(2N)$. Soft thresholding gives, for $1 \leq k \leq N$,

$$x^k = w^k = R - (k-1)h, \qquad y^k = w^{k+1} = R - kh.$$

The positive branch is valid because $y^k \geq R/2 > 0$. Therefore $y^N = R/2$ and

$$F(y^N) - F(0) = \frac{R^2}{4\gamma N}, \qquad |w^1 - w^\star| = R.$$

The zero function satisfies any prescribed smoothness upper bound $L > 0$. If the minimal Lipschitz constant is required to equal L, use

$$f(u, v) = \tfrac{L}{2}v^2, \qquad g(u, v) = \frac{R|u|}{2\gamma N}, \qquad w^1 = (R, 0).$$

The second coordinate stays zero and the first follows the same trajectory. Both functions are finite-valued. This completes the sharpness statement in Theorem 1.

5 An operator residual counterexample

For maximally monotone operators A, B on a finite-dimensional Euclidean space, the Douglas–Rachford iteration in [1, Algorithm 1] is

$$\begin{aligned} x^k &= (I+\gamma B)^{-1} w^k, \\ y^k &= (I+\gamma A)^{-1}(2x^k - w^k), \\ w^{k+1} &= w^k + \lambda(y^k - x^k) = Tw^k, \qquad k = 1, \dots, N. \end{aligned} \tag{20}$$

The initial state is w^1. In particular, the residual $Tw^N - w^N$ equals $w^{N+1} - w^N$, after $N-1$ preceding state updates.

A single-valued operator B is β-cocoercive if

$$\langle Bx - By, x - y\rangle \geq \beta \|Bx - By\|^2 \qquad (x, y \in \mathbb{R}^n).$$

Conjecture 1 of the version dated 13 September 2025 [1] assumes that T has a fixed point $w^\star$, that B is β-cocoercive, and that $A = \partial g$ for a closed proper convex function g. It proposes, for $0 < \gamma < \beta$ and $0 < \lambda < 2$,

$$\|Tw^N - w^N\|^2 \leq \frac{\lambda^2}{((N-1)\lambda + 1)^2} \|w^1 - w^\star\|^2. \tag{21}$$

This is PDF Conjecture 1, or Conjecture 3.5 in the HTML rendering. The stated assumptions do not require B to be a gradient. We show that this distinction matters for (21).

The construction has classical antecedents. Planar damped rotations provide finite-horizon residual lower bounds in proximal-point and averaged-operator analysis; see Lieder [6, Section 4] and Gu–Yang [7, Section 5]. Morin–Giselsson [8, Section 4.1] explicitly use a cocoercive averaged rotation to distinguish general operators from gradients. The averagedness of a cocoercive resolvent is also standard [9, Proposition 5.2]. We apply this established mechanism to the precise claim (21), without asserting a new general lower-bound technique or an optimal constant for the full splitting class.

5.1 An exact rational example

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A = 0 = \partial g \quad (g \equiv 0), \qquad B = \frac{I + 4J}{17}.$$

Fix

$$\beta = 1, \qquad \gamma = \frac{1}{2}, \qquad \lambda = 1, \qquad w^1 = (1, 0), \qquad w^\star = 0. \tag{22}$$

Proposition 5.1. The operators and parameters above satisfy every stated hypothesis of (21). Their actual recurrence (20) satisfies

$$\|Tw^N - w^N\|^2 = \frac{68^{N-1}}{73^N} \qquad (N \geq 1). \tag{23}$$

At $N = 14$ this exceeds the conjectured upper bound $1/14^2$.

Proof. The zero map is the subdifferential of the closed proper convex function $g \equiv 0$. Also

$$B^T B = \frac{1}{17} I, \qquad \frac{B + B^T}{2} = \frac{1}{17} I.$$

Thus $\langle Bh, h\rangle = \|Bh\|^2$ for every $h \in \mathbb{R}^2$, proving 1-cocoercivity and monotonicity.

Any everywhere-defined linear monotone map C is maximally monotone in this setting. Indeed, if (z, v) is monotonically related to its graph, substituting $z + th$ gives

$$-t\langle v - Cz, h\rangle + t^2\langle Ch, h\rangle \geq 0 \qquad (t \in \mathbb{R}).$$

Taking arbitrarily small t of both signs forces $v = Cz$. This applies to A and B.

Since $(I + \gamma A)^{-1} = I$ and $\lambda = 1$, the actual substeps in (20) give $y^k = 2x^k - w^k$ and $w^{k+1} = x^k$. Direct inversion yields

$$T = (I + B/2)^{-1} = \frac{1}{73}\begin{pmatrix} 70 & 8 \\ -8 & 70 \end{pmatrix}. \tag{24}$$

Thus 0 is a fixed point and the initial squared distance is one. The strict parameter ranges follow from (22). Finally,

$$T^T T = \frac{68}{73} I, \qquad (T - I)^T (T - I) = \frac{1}{73} I. \tag{25}$$

As $w^N = T^{N-1} w^1$, these identities prove (23). The strict violation at horizon fourteen is the exact integer inequality

$$196 \cdot 68^{13} - 73^{14} = 823324207703713330771591 9 > 0.$$

No numerical approximation is needed. ∎

The matrix (24) has contraction factor $\sqrt{68/73} < 1$. This fixed example therefore converges geometrically, while violating the particular finite-horizon estimate (21).

6 A family explaining the order obstruction

The failure is not confined to the constant in (21). Keep $A = 0$, $\beta = 1$, $\gamma = 1/2$, $\lambda = 1$ and $w^1 = (1, 0)$. For each integer $N \geq 2$, set

$$a_N = \frac{4}{5(N-1)}, \qquad B_N = a_N I + \sqrt{a_N - a_N^2}\, J. \tag{26}$$

The subscript N labels a different problem instance for each horizon; within that instance, the same operator is used at every iteration.

Proposition 6.1. Every operator pair in (26) satisfies the hypotheses of (21). For its actual DR map $T_N = (I + B_N/2)^{-1}$,

$$\|T_N w^N - w^N\|^2 = \frac{(N-1)^{N-1}}{5N^N}. \tag{27}$$

Consequently, the class in the stated conjecture admits no uniform squared-residual bound of order $O(N^{-2})$ in terms of the initial squared distance alone.

Proof. For $N \geq 2$, one has $0 < a_N < 1$. More generally, let $0 < a < 1$ and $b = \sqrt{a - a^2}$. The linear operator $B = aI + bJ$ satisfies

$$B^T B = (a^2 + b^2)I = aI, \qquad \langle Bh, h\rangle = a\|h\|^2 = \|Bh\|^2.$$

It is 1-cocoercive and maximally monotone by the argument above. The zero map remains a convex subdifferential, and 0 is a DR fixed point.

Write $T = (I + B/2)^{-1}$. Since $J^T J = I$ and $J^T = -J$, direct multiplication gives

$$(I + B/2)^T (I + B/2) = \left(1 + \frac{5}{4}a\right) I.$$

Also $T - I = -\frac{1}{2}BT$. Hence

$$\|Tw^N - w^N\|^2 = \frac{a/4}{(1 + 5a/4)^N} \qquad \text{when } \|w^1\| = 1. \tag{28}$$

Substituting $a = a_N$ proves (27), since $1 + 5a_N/4 = N/(N-1)$. Finally,

$$N\|T_N w^N - w^N\|^2 = \frac{1}{5}\left(1 - \frac{1}{N}\right)^{N-1} \longrightarrow \frac{1}{5e} > 0.$$

Thus multiplying these squared residuals by N^2 gives an unbounded sequence despite a common unit initial distance and common parameters. This excludes a uniform $O(N^{-2})$ bound. ∎

Each individual T_N is a strict contraction. Formula (27) concerns the worst-case class as the horizon varies, not a sublinear asymptotic rate for one fixed operator. It supplies a lower bound only; the factor $1/5$ is not asserted to be globally optimal over all admissible A, B.

For clarity about ancestry, the family $aI + \sqrt{a - a^2}J$ can be written as $(I + R_\theta)/2$, where R_θ is a planar rotation with $\cos\theta = 2a - 1$. This is the standard cocoercive averaged-rotation family used in [8]. Its resolvent in the present normalization is 1/6-averaged, consistent with the known transformation in [9]. Lieder's sharp global KM theorem [6, Theorem 4.9] is stated for averaging at least $1/2$; it is not being invoked as a global sharpness theorem at $1/6$.

The nonzero skew part makes these B nonsymmetric and hence not gradients of convex functions. Conjecture 2 in [1] concerns composite convex optimization and requires the first-resolvent operator B to be a convex gradient. The examples above address Conjecture 1 as stated. They do not apply to the convex-gradient class. Theorem 1 establishes the unrelaxed small-step part of the separate objective-gap conjecture; its remaining parameter range and any gradient-restricted version of (21) are not resolved here.

A The interior row polynomial

The polynomial in (18) is the following explicit positive-coefficient expression, for $u, q \geq 0$:

$$\begin{aligned} Q(u,q) = {} & 2u^8 + (18q+66)u^7 + (64q^2+489q+912)u^6 \\ & + (112q^3+1366q^2+5366q+6820)u^5 \\ & + (96q^4+1740q^3+11139q^2+30267q+29684)u^4 \\ & + (32q^5+936q^4+9380q^3+42880q^2+91878q+74848)u^3 \\ & + (128q^5+2604q^4+20388q^3+77207q^2+141824q+101228)u^2 \\ & + (128q^5+2274q^4+15854q^3+54138q^2+90376q+58856)u \\ & + 4q^5+120q^4+1064q^3+4080q^2+7072q+4480. \end{aligned} \tag{29}$$

For a direct derivation that avoids expanding a large matrix, let

$$s_j = 2j - \beta_j - \beta_{j-1} \quad (j < N), \qquad b_j = 2A_j + 2E_{j-1} + \sigma_j \quad (j < N).$$

The row sum from (13), for $2 \leq i \leq N-2$, is exactly

$$\begin{aligned} \eta_i = {} & b_i - E_i - \tfrac{1}{2}\sigma_i - E_{i-1} - \tfrac{1}{2}\sigma_{i-1} \\ & - \tfrac{1}{4}\left(\frac{2(i-1)s_{i-1}}{\kappa_{i-1}} - \frac{\beta_i s_i}{\kappa_i} - \frac{\beta_i s_{i+1}}{\kappa_{i+1}} \right). \end{aligned} \tag{30}$$

Multiply by the denominator in (18), substitute $i = u+2$ and $N = u+q+4$, and collect powers of u. The result is (29). This is an identity of rational functions in the indeterminate horizon and index. Since $u = i-2 \geq 0$ and $q = N-i-2 \geq 0$ exactly describe the interior range, it proves the sign over the whole range. At the penultimate row, the last form in (12) has row sum $s_N = 2N/(N+1)$; substituting this terminal value gives (17). The first and last rows follow directly from the same coefficient forms.

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